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# Performance evaluation of methanogenic digester using kitchen waste for validation of optimized hydrolysis conditions for reduction in ammonia accumulation

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ABSTRACT

Ammonia accumulation above the inhibitory level is the major cause of failure of anaerobic digesters (AD) treating kitchen waste (KW). In the present investigation a biphasic system (hydrolysis and methanogenesis) was evaluated for validation of previously optimized hydrolysis conditions [pH 7.2–7.5, Aeration 0.19–0.22 vvm (volume of air per unit volume of media per minute), Temperature 44 - 48 °C] for reduction in ammonia accumulation in the methanogenic digester. The pre hydrolyzed KW was fed to methanogenic reactor with the organic loading 1.16 gVS/L - 2.97 gVS/L. Specific biogas and methane yield were 0.589 L biogas/$gVS_{feed}$ and 0.321 L methane/$gVS_{feed}$ under mesophilic condition, and 0.633 L biogas/$gVS_{feed}$ and 0.329 L methane/$gVS_{feed}$ under thermophilic condition, respectively. The ammonia concentration in the methanogenic phase remained below 250 ppm and 550 ppm under mesophilic and thermophilic conditions, respectively, validate the optimized conditions for hydrolysis of KW for reduction in ammonia accumulation in methanogenesis phase. The Microbial diversity analysis of methanogenic phase was carried out to monitor the shifting of microbial population under mesophilic and thermophilic conditions. Results on these aspects have also been presented and discussed in this paper.

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## 1. Introduction

Municipal solid waste (MSW) includes waste from households, commercial and institutional establishments, markets and street sweepings [1]. Safe and cost-effective management of organic fraction of MSW (OFMSW), such as kitchen waste (KW) is a significant environmental challenge for modern society. Inadequately managed KW disposal has the potential to affect the health and environment [2]. KW is a readily available organic-rich waste, and has great potential for anaerobic digestion [3]. Conversion of KW into biogas using anaerobic digestion is one of the promising approaches for solving the OFMSW management problem, since, it also provides clean energy in the form of methane [4].

Despite the advantages of biogas production through anaerobic digestion of KW, accumulation of by-products viz. ammonia, hydrogen sulfide and long chain fatty acids has been well known for inhibiting the process of biomethanation. Ammonia accumulation in the biogas digester treating substrate having low C/N ratio has been reported as a major cause of digester failure. However, there is a large variation in the reported inhibitory ammonia concentration (1.7–14 g/L) which depends on the inoculum, substrate, inoculum acclimatization and environmental conditions (pH and temperature) [5,6]. Kayhanian [7] explained the theoretical relationship between the concentrations of free ammonia and total ammonia, pH and temperature dependent dissociation constant, and concluded that in thermophilic bioreactor system free ammonia concentration is six times greater than mesophilic system. The most widely accepted mechanism explaining ammonia inhibition is that elevated ammonia level causes change of intracellular pH, increase in maintenance energy requirement, depletion of intracellular potassium, and inhibition of enzyme reactions involved in methanogenesis [8]. Accumulation of ammonia in the digester causes shifting in the biochemical pathways towards syntrophic acetate oxidation from aceticlastic methanogenesis [9].

The entire process of biomethanation can be carried out either in a monophasic system, where all the four steps are carried out in a single digester, or in a biphasic system, in which the fermentative

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(hydrolysis and acidogenesis) phase is separated from the methanogenic phase. As compared to monophasic system, the biphasic system could be controlled more easily and efficiently [10]. Since ammonia is produced during the hydrolysis and acidogenesis phase only, therefore, fermentation of KW can be optimized to control and remove most of the ammonia from hydrolysate prior to feed in the methanogenic phase.

In our previous study, the aerobic hydrolysis of KW was studied and statistically optimized for simultaneous solubilization and ammonia removal by controlling pH, temperature and aeration rate [11]. In the present study, we have described the performance of a methanogenic digester fed with hydrolyzed KW, which was generated under the optimized conditions. The performance of methanogenic phase of a biphasic semi-continuous biogas digester (substrate fed once in a day), and changes in the microbial diversity were monitored. Results were compared and discussed for methane production, ammonia concentration and metagenomic analysis for microbial diversity during the biogas digester operation under mesophilic and thermophilic conditions.

## 2. Material and method

### 2.1. Kitchen waste processing and hydrolysis

KW was collected from different locations (institutional canteen, guest house and research scholar hostel) of CSIR-National Environmental Engineering Research Institute, Nagpur, India. The KW mainly consisted of cooked rice, vegetable curries, pulse soup and chapati. The collected KW was mixed and grinded to make a uniform slurry. The KW slurry was characterized for total solids (TS) and volatile solids (VS) using standard protocol as described in standard methods for examination of water and waste water [12].

Hydrolysis of KW slurry was carried out separately in a fermenter of 3 L capacity under previously optimized conditions [pH 7.2–7.5, temperature 43–48 °C, aeration 0.19–0.22 vvm (volume of air per unit volume of media per minute)] for a retention time of 24 h [11]. The hydrolyzed KW (hydrolysate) having $0.218 \pm 0.003$ gTS/mL, $0.213 \pm 0.005$ gVS/mL, 1419 mg/L VFA, $154 \pm 31$ mg/L TAN, $55 \pm 10$ mg/L FAN was fed once in a day to the methanogenic chamber for biogas production as shown in Table 1.

**Table 1**
Organic loading rates during the reactor operation under mesophilic and thermophilic conditions.

| Days | Temperature | OLR (gVS/L/D) | SRT (D) | g food (w/v) |
|---|---|---|---|---|
| 1–18 | 37°C | 1.14 | 51 | 173.33 g/700 ml |
| 19–41 | | 1.48 | 51 | 225 g/700 ml |
| 42–77 | | 1.68 | 45 | 255 g/800 ml |
| 78–120 | | 1.68 | 40 | 255 g/900 ml |
| 1–12 | 46°C | 1.14 | 73 | 173 g/500 ml |
| 12–26 | | 1.69 | 51 | 260 g/700 ml |
| 27–49 | | 1.69 | 45 | 260 g/800 ml |
| 50–84 | | 1.84 | 40 | 283 g/900 ml |
| 85–98 | | 1.95 | 36 | 300 g/800 ml |
| 99–115 | | 2.6 | 30 | 400 g/1000 ml |
| 116–150 | | 2.98 | 30 | 458 g/1000 mL |

### 2.2. Biogas digester and operation

A lab-scale digester (methanogenic chamber) of 36 L working volume and 24 L head space volume was fabricated and used for anaerobic digestion of pre-hydrolyzed KW. The digester consisted of inlet for feeding the hydrolyzed slurry and an outlet for the digested slurry. The digester was equipped with a pH and temperature probe to continuously monitor the parameters during the process. The temperature of digester was controlled by a circulating water jacket and agitator used for uniform heat transfer. The inoculum used for starting-up the digester was collected from a running local mesophilic biogas plant, fed with cow dung. 36 L inoculum having $0.078 \pm 0.001$ gTS/mL and $0.057 \pm 0.004$ gVS/mL was filled in the digester. The KW slurry (previously hydrolyzed under optimized conditions) was fed into digester by increasing organic loading from 1.14 gVS/L to 1.68 gVS/L under mesophilic condition (37 °C). Next, the temperature of methanogenic phase was gradually increased by 1 °C after every 3 days up to reaching 46 °C to prevent heat shock during acclimatization of inoculum for thermophilic condition at constant OLR (1.14 gVS/L). After attaining 46 °C temperature, the organic loading was gradually increased from 1.69 gVS/L to 2.98 gVS/L as shown in Table 1.

The performance of biogas digester fed with the optimally hydrolyzed KW, in terms of methane yield and other important parameters (ammonia, Volatile fatty acids and alkalinity) was evaluated under mesophilic and thermophilic conditions. Quantitative analysis of biogas produced was done daily by measuring head space pressure using a pressure gauge. After taking the reading, head space overpressure was released, so that the starting pressure was returned to zero. The volume of biogas generated at STP was calculated using following equations:

$$n = \frac{P*V}{R*T}. \quad (1)$$

$$v = n \times 22.4. \quad (2)$$

where, n = mole of biogas; P= Head space pressure (bar); V= Head Space volume of digester (Liter); R = gas constant (bar $Lmole^{-1}K^{-1}$); T = temperature (K); v = volume of biogas (Liter).

The quality of biogas was checked out by monitoring the methane content in GC-TCD (Agilent) equipped with a stainless steel Porapak Q column (column length 6 ft and ID 2 mm). The gas samples from the biogas digester were collected in the 1 L Tedlar bags. 100 μL of gas sample was injected in the injector port of GC-TCD with the help of gas-tight syringe (Hamilton, USA). The run time of analysis was preset for 2 min with the column temperature of 50 °C. The temperature of injector, oven, and detector was 100 °C, 60 °C and 100 °C, respectively. Hydrogen was used as the carrier gas. The result of the analysis is based on the principle that the gas having a thermal conductivity greater then hydrogen shows the peak at specific retention time. The peak area (Volts) of graph corresponded to the % concentration of methane, carbon dioxide and nitrogen in the biogas sample.

Other indicators of process stability viz. pH, alkalinity, VFA, ratio of VFA to alkalinity and ammonia concentration were monitored once a week using standard protocol as described in Standard methods for the examination of water and wastewater [12].

### 2.3. Microbial diversity analysis

#### 2.3.1. DNA extraction

Microbial diversity analysis of inoculum and sample collected on 54th day of digester operation under mesophilic condition and 75th day of biogas digester operation under thermophilic condition

was done. Samples were designated as SM-EBD for inoculum, MP-EBD for mesophilic condition sample and TP-EBD for thermophilic condition sample. All samples were stored at −20 °C until the extraction of DNA. Total genomic DNA was extracted using the Fast DNA Spin kit for soil (MP Biomedicals, Santa Ana, CA, USA) according to the manufacturer's instructions. The concentrations and quality of extracted DNA were measured using a Nanodrop spectrophotometer (Nanodrop, ND-8000).

*2.3.2. Metagenome sequencing*

Qubit fluorometery was used to determine the DNA concentration and purity of samples. Illumina-compatible sequencing library preparation for DNA samples was performed using NEXTFlex DNA Sequencing Kit (Bioo Scientific) according to manufacturer instructions. In brief, DNA shearing was performed to generate fragments of approximately 200bp - 400bp size using CovarisS220 system (Life Technologies, USA). The fragment size distribution was confirmed on Agilent TapeStation with D1000 DNA Kit (Agilent, Germany). Further, the sheared DNA was purified using HighPrep beads (Magbio Genomics, USA) followed by end repair, A-tailing, and ligation with the Illumina multiplexing adapters (Bioo Scientific, Austin) according to the protocol outlined in NEXTFlex DNA Sequencing kit. The ligated DNA was purified using HighPrep beads and further subjected to 10 rounds of PCR (denaturation at 98 °C for 2 min, cycling at 98 °C for 30 sec, 65 °C for 30 sec and 72 °C for 60 sec, and final extension at 72 °C for 4 min) with Illumina compatible primers provided within the kit. The resultant PCR product (Illumina-compatible sequencing library) was purified using HighPrep beads, quantified using Qubit fluorometer and fragment size distribution analysis was performed on AgilentD1000 TapeStation. KAPAqPCR quantification kit (KAPA Biosystems; Massachusetts) was used to quantify the Illumina-compatible sequencing library prior to loading on Illumina Sequencer flowcell. Adapter details:

Universal Adapter: 5′AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT 3' and Index Adapter: 5′GATCGGAAGAGCACACGTCTGAACTCCAGTCAC [INDEX] ATCTCGTATGCCGTCTTCTGCTTG 3'.

*2.3.3. Bioinformatics analysis*

Phylogenetic and functional diversity of samples were analyzed using MG-RAST (Metagenomics Rapid Annotation using Subsystem Technology) server by submitting the assembled reads using online account. For the analysis default parameters which included the maximum e-value cut-off ($10^5$), minimum percent identity cut-off (60%), minimum alignment length cut-off (15 bps) and minimum abundance cut-off (1) were selected for filtering the data from the data selection menu. Phylogenetic and functional analysis of data was done using REFSeq and Subsystems database.

**Accession number**: The metagenomic reads have been submitted under the study accession number PRJNA357876 which includes the read files for MP-EBD: SRS1925843, TP-EBD: SRS192581 and SM-EBD: SRS1925844.

## 3. Result and discussion

*3.1. Performance of biogas digester*

The kitchen waste hydrolyzed under optimum conditions was fed to the biogas digester for methanogenesis. The target of hydrolysis optimization was to minimize ammonia accumulation in the biogas digester. The methanogenic digester was initially operated under mesophilic conditions (37 °C) for 120 days and then under thermophilic conditions (46 °C) for 150 days.

Fig. 1a and b show the biogas yields at different organic loading rates (OLR) during mesophilic and thermophilic operations. The mesophilic operation was started with an organic loading rate of 1.14 gVS/L which was gradually increased up to 1.68 gVS/L. Under this condition, average daily biogas production was 17, 27 and 42 L at organic loading rates of 1.14, 1.48 and 1.68 gVS/L, respectively. The thermophilic operation was started at OLR of 1.14 gVS/L which was gradually increased up to 2.97 gVS/L. Under this condition, average daily biogas production was 37, 46, 45 and 63 L at organic loading rates of 1.14, 1.69, 1.84 and 2.6 gVS/L, respectively, which are higher than the mesophilic operation. The average daily methane production was also higher in thermophilic operation, at similar OLR as shown in Fig. 1a and b.

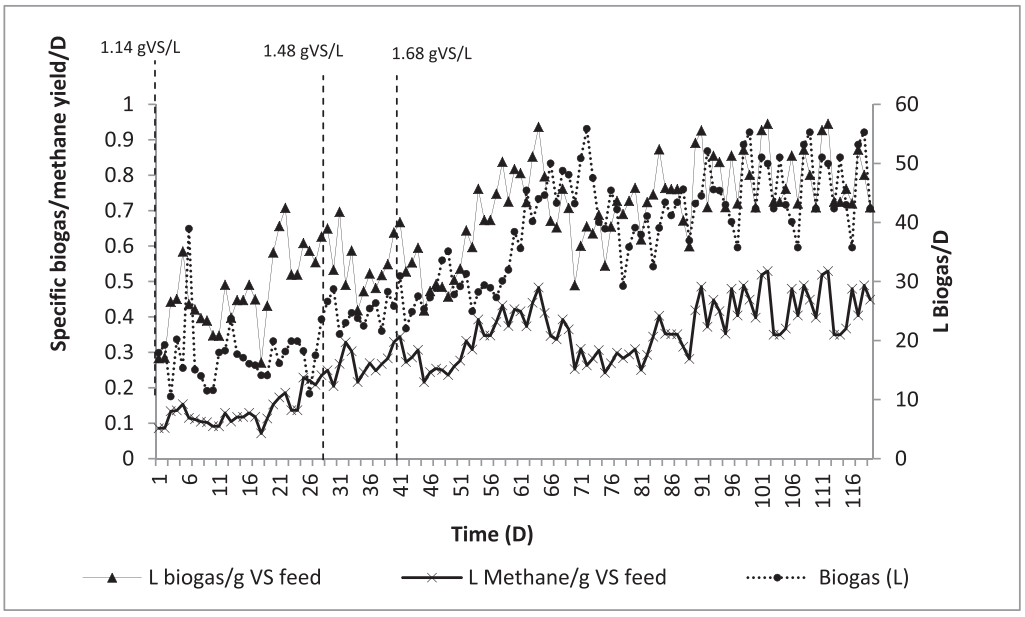
The methane content of the biogas also increased from 36% to 54% upon an increase in the organic loading from 1.14 to 1.68 gVS/L for mesophilic operation. In the case of thermophilic operation, the methane content increased from 43% to 55% upon an increase in the organic loading from 1.14 to 2.97 gVS/L (Fig. 2).

After reaching stable performance, the Specific biogas and methane yields were 0.589 L biogas/$gVS_{feed}$ and 0.321 L methane/$gVS_{feed}$ under mesophilic condition, and 0.633 L biogas/$gVS_{feed}$ and 0.329 L methane/$gVS_{feed}$ under thermophilic conditions respectively. The average methane content in biogas produced was 48% and 50% under mesophilic and thermophilic conditions respectively. This shows the thermophilic operation of biogas digester is more efficient for higher methane recovery at similar organic loading rates.

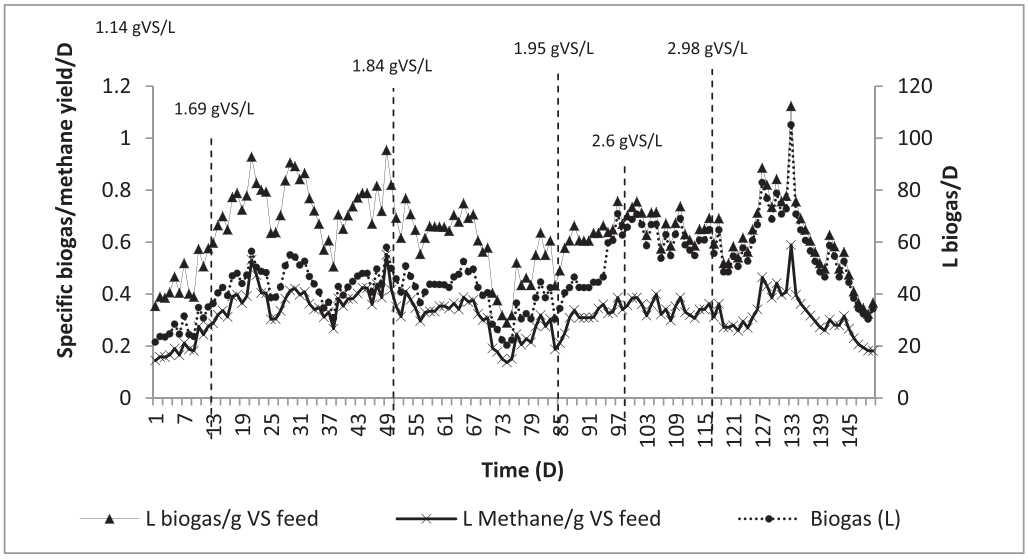
Fig. 3a–b shows free ammonia concentration in the biogas digester slurry. During mesophilic operation, the free ammonia concentration in the digester remained below 220 mg/L during the 120 days period. During thermophilic operation, the free ammonia concentration was higher and remained below 550 ppm during 150 days of operation. Higher free ammonia concentration in case of thermophilic operation could be explained on the basis of a relationship provided by Zhang et al. [13]; Garcia and Angenent [14] and Strik et al. [15], according to which the higher temperature or the higher pH, the higher amount of free ammonia produced.

Consequently, at high temperature, it is suitable to operate biogas digester at lowest possible pH. However in our study, both thermophilic and methanogenic digestions were operated at similar pH (7.25–7.5), therefore higher ammonia accumulation was observed in the case of thermophilic operation. Nevertheless, in both the cases of this investigation, the ammonia concentration in the digestion slurry was much lesser than that reported earlier in the literature. Benabdallah El Hadj et al. [16] reported that the methane generation under mesophilic and thermophilic conditions was reduced by 50% when ammonium ion reached concentrations of 3,860 and 5,600 mg $NH_4$-N/L respectively at pH 7–8.5 [17]. reported that mesophilic digester had significantly lower ammonium ion concentration (640–800 mg/L $NH_4$–N) than those in the thermophilic digester (2880–3840 mg/L $NH_4$–N). Kayhanian [7] also showed that free ammonia nitrogen concentration in a thermophilic biogas digester can be expected to be 6 times higher than in a mesophilic digester at the same pH. Although undesirable for methanogenesis, the ammonia concentration found in the present study is a good indicator of the bacterial activity since ammonia is one of the end products of microbial hydrolysis of proteins and higher ammonia concentration could be associated to higher protein degradation due to the faster kinetics of the biological activity.

VFA, alkalinity and their ratio have been considered as important indicators of stability of biogas digester. The measured concentration of VFA and alkalinity and their ratio is showed in Fig. 4a–b. During mesophilic operation, the VFA was found to be highly varied and reached more than 10,000 mg/L (Fig. 4a). In case of thermophilic operation, the VFA was found to be much lower, between 1500 and 2000 mg/L (Fig. 4b). The alkalinity was also




(a)




(b)

**Fig. 1.** Graphs showing specific biogas and methane yield on primary axis and L biogas produced per day on secondary axis under mesophilic condition (a) and thermophilic condition (b). Lines with dashes showing the organic loading.

found to be higher during mesophilic compared to the thermophilic operation and there was a gradual rise in alkalinity with increase in OLR. This could be due to the increasing load of sodium hydroxide used for maintaining the pH during hydrolysis of kitchen waste under optimized conditions. During most of the operation period, the VFA to alkalinity ratio was below 0.3 in both the cases, which has been reported optimum for a robust biomethanation process [18]. Lower ammonia accumulation, low VFA/alkalinity ratio with high methane production in the biogas digester shows that the hydrolyzed kitchen waste was suitable for methanogenesis and hence validates the optimized conditions of kitchen waste hydrolysis as discussed in section 3.1.

### 3.2. Metagenome analysis

Inoculum sample (SM-EBD), and samples collected on 54th and 75th day of biogas digester operated under mesophilic (MP-EBD) and thermophilic conditions (TP-EBD) were analyzed for microbial diversity. Extracted DNA libraries were in the expected size range of 300bp – 700bp on the TapeStation. The effective sequencing insert size is 180bp – 580bp, with combined adapter size being ~120bp. The library was found suitable for sequencing on Illumina platform.

NGS-based metagenomic approach was applied to follow the development of the microbial community structure in the biogas reactors at different temperature conditions. Striking reorganization of the mesophilic microbial community into a stable thermophilic community without any inoculum addition was observed in

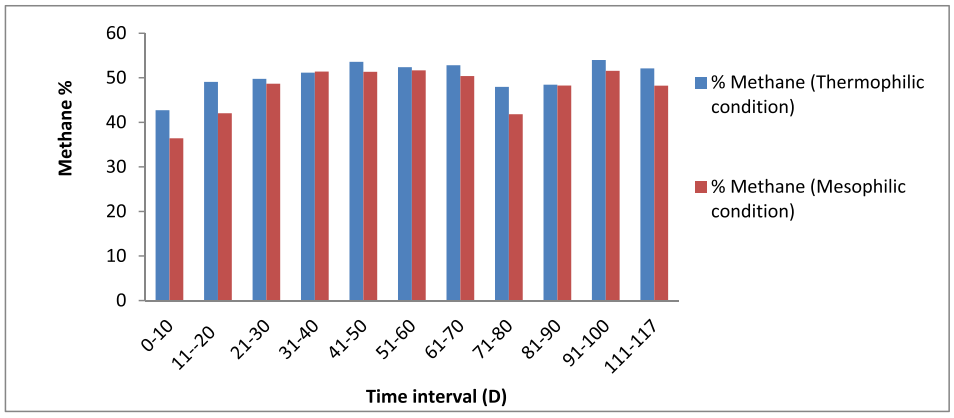
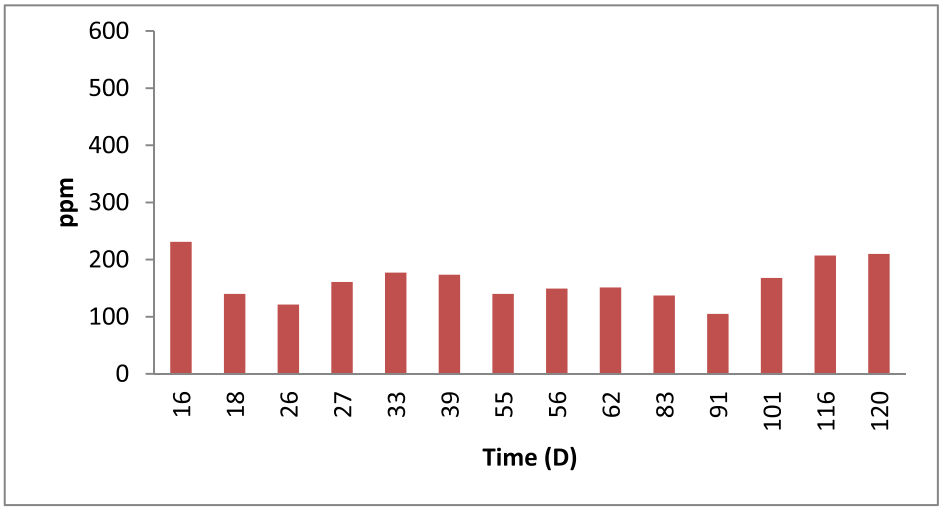
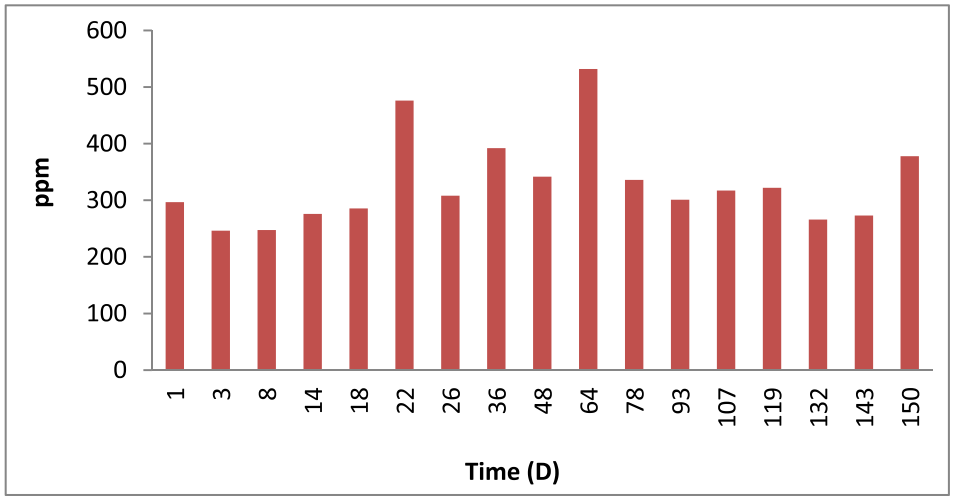



**Fig. 2.** Methane % during Thermophilic and Mesophilic operation of the biogas digester.

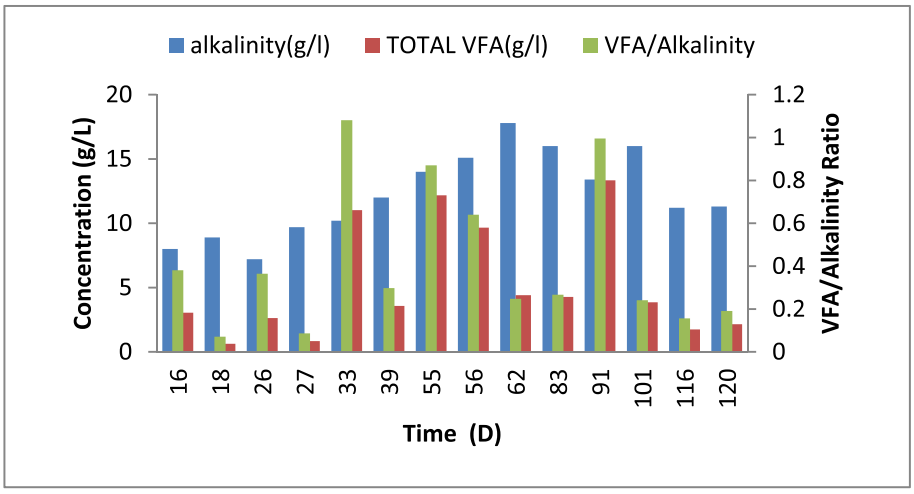



(a)

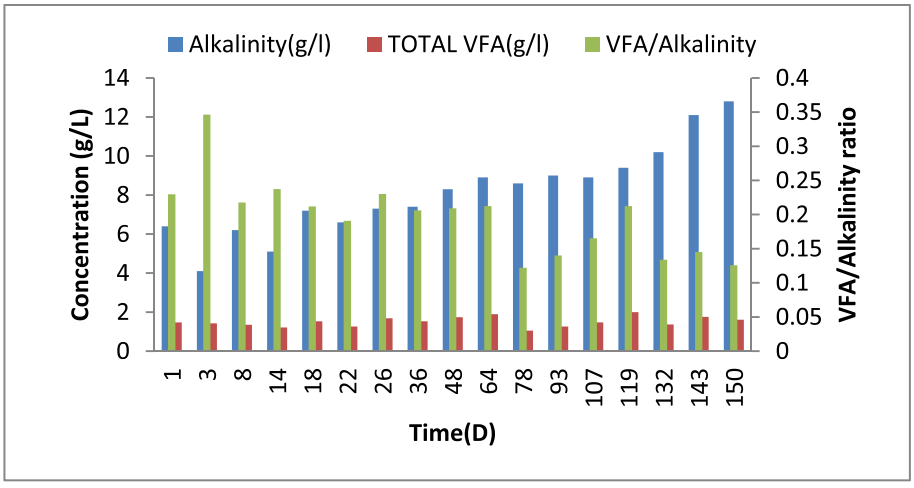



(b)

**Fig. 3.** Free ammonia concentration under (a) mesophilic and (b) thermophilic operation of the biogas digester.




(a)




(b)

**Fig. 4.** Alkalinity, total VFA and VFA to Alkalinity ration under (a) mesophilic and (b) thermophilic operation of biogas digester.

the anaerobic digester in response to temperature elevation which means re-organization was directed by the increasing temperature as selection pressure [19–21]. Community analysis revealed that the general diversity of the inoculum was higher than that of the mesophilic and thermophilic samples [21]. Noticeable changes were observed in the microbial diversity during the whole anaerobic digestion period, shown in Table 2.

Reduction of bacterial count in the thermophilic condition showed that temperature is a selection measure of microbial diversity. At order level majority of the genera (89.1%) were common to all 3 metagenome datasets. About 2–6 bacterial orders were unique to or shared among 2 out of 3 samples (Fig. 5a).

Hence overall the community did not distinctly differ, but there were remarkable changes in the abundance of the microbial members in three samples. Similarly at genera level majority of features were common to all 3 datasets. Sample MP-EBD contained few unique genera, 50% of which were commonly shared with SM-EBD sample, and 50% of which were included exclusively in MP-EBD sample as observed in Venn analysis (Fig. 5b). The list of features categorized unique and shared across three metagenome are enlisted in supplementary Table x1 and y1.

**Table 2**
% Abundance at Domain level in inoculum (SM-EBD), biogas digester sample operated under mesophilic condition (MP-EBD) and under thermophilic condition TP-EBD.

| Domain | SM-EBD | MP-EBD | TP-EBD |
|---|---|---|---|
| Archaea | 6.4 | 7.0 | 8.1 |
| Bacteria | 92.7 | 92.2 | 91.1 |
| Eukaryota | 0.9 | 0.7 | 0.7 |
| Viruses | 0.1 | 0.1 | 0.0 |
| Other sequences | 0.0 | 0.0 | 0.0 |

### 3.2.1. Taxonomic diversity

*3.2.1.1. Phylum abundance.* Complete restructuring was detected in the phylum abundance. In sample SM-EBD the major bacterial components of the digestion ecosystem were the phylum *Proteobacteria* (25.9% of total bacteria), *Bacteroidetes* (24.4% of total bacteria), *Firmicutes* (15% of total bacteria), unclassified (derived from bacteria) (6.3% of total bacteria), *Euryarchaeota* (5.8% of total archaea), *Actinobacteria* (3% of total bacteria), *Spirochaetes* (3% of

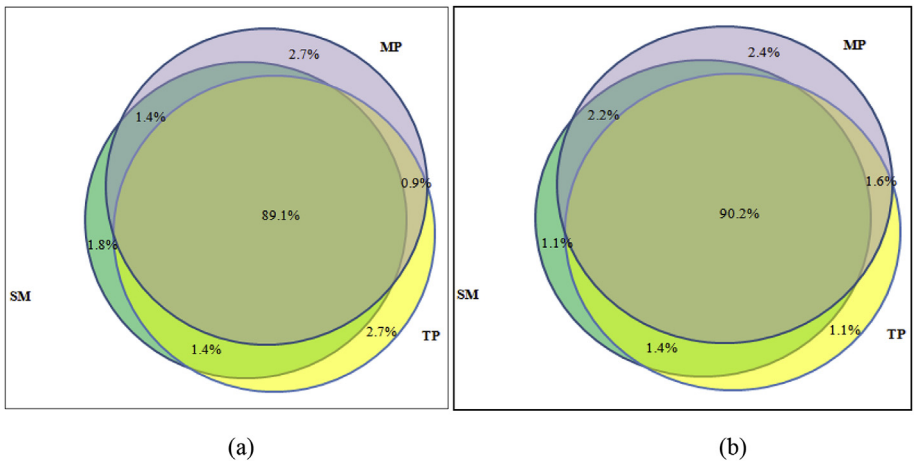



**Fig. 5.** Venn analysis of unique to or shared bacterial (a) orders and (b) genera among inoculum: SM-EBD (SM), biogas digester sample operated under mesophilic condition: MP-EBD (MP) and under thermophilic condition: TP-EBD (TP).

total bacteria), *Acidobacteria* (2.1% of total bacteria) and *Chlorobi* (2% of total bacteria). Similarly, in the MP-EBD and TP-EBD samples *Firmicutes, Bacteroidetes* and *proteobacteria* were the major bacterial components which were 20.2%, 33.3%, and 13.5% in MP-EBD and 16.1%, 17.6% and 21.4% in TP-EBD samples respectively (Fig. 6). The abundance of the phylum *Proteobacteria*, *Firmicutes* and *Bacteriodetes* in the biogas digester has been reported previously. In MP-EBD other dominating phylum were *Actinobacteria* (7.7%), *Euryarchaeota* (6.8%) *Synergistetes* (2.5%) and unclassified (derived from bacteria, 2.1%) while in the TP-EBD other dominating phylum were *Thermotogae* (15.2%), *Euryarchaeota* (7.8%), *Actinobacteria* (2.9%), *Planctomycetes* (2.7%) and unclassified from bacteria (2.6% of total bacteria). The abundance of phylum *Thermotogae* in TP-EBD (15.2%) was extremely higher than the SM-EBD and MP-EBD samples (0.93% and 2.9%, respectively). The changes in abundance of these microbial communities is subject to selection of optimum growth temperature, for example, *Actinobacteria* requires low temperature for their growth therefore, showed higher abundance at mesophilic condition and increased temperature caused reduction of their abundance at thermophilic condition and vice-versa is applicable for *Thermotogae*.

*3.2.1.2. Genera abundance. Bacteroides* (6.3%), *Candidatus Cloacamonas* (6.2%), *Clostridium* (3.3%), *Geobacter* (2.1%), *Prevotella* (1.7%), *Parabacteroides* (1.6%) *Syntrophus* (1.4%) and *Treponema* (1.3%) were the dominated genera present in the inoculums (Fig. 7a). *Candidatus Cloacamonas acidaminovorans* has been reported for their high abundance in several anaerobic digester operated under mesophilic condition for anaerobic digestion of high C : N containing substrate viz. maize silage [22,23].

At mesophilic condition, microbial abundance changed during acclimatization of inoculum for kitchen waste. *Bacteroides* (10%), *Clostridium* (5.5%), *Parabacteroides* (2.4%), *Prevotella* (2.35%), *Candidatus Cloacomonas* (2%), *Pedobacter* (1.4%) present in MP-EBD as major genera showed the breakdown and further decomposition of kitchen waste at mesophilic condition (Fig. 7b). The presence of *Bacteroides* (7%), *Clostridium* (3.3%), *Parabacteroides* (1.7%), *Prevotella* (1.3%), *Pedobacter* (0.5%), *Treponema* (0.2%) at lower frequency in TP-EBD showed that these are temperature sensitive/tolerant microbes (Fig. 7c). In thermophilic condition, *Candidatus cloacamonas*, *Geobacter*, *Syntrophus, Anaeromyxobacter, Bacillus, Desulfovibrio, CandidatusSolibacter* were the dominated genera involved in the breakdown and fermentation of organic molecules. The increased abundance of these genera at thermophilic condition showed that these microbial communities are temperature tolerant and can grow up to 50 °C.

In the thermophilic sample *Bacteroides* (7.4%), *Methanoculleus* (6.0%), unclassified genera derived from *Thermotagales* family (5.3%), *Clostridium* (2.8%), *Kosmotoga* (2.6%), *Geobacter* (2.1%) and *Thermotoga* (1.8%) were the dominated genera overall. The abundance of *Candidatus Cloacamonas acidaminovorans* was 6.2% in SM-EBD which was drastically reduced in the mesophilic (2%) and thermophilic sample (2.5%). Proteome analysis indicated that *C. acidaminovorans* might gain energy from sugars in the Embden-Meyer hof pathway and from the fermentation of amino acids, and thereby produces hydrogen and carbon dioxide [22].

The abundance of *Methanoculleus* (4.3%), *Kosmotoga* (3.4%), *Thermotoga* (2.2%) were higher in TP-EBD than the SM-EBD (0.5%, 0.1%, 0.4% respectively) and the MP-EBD (1.7%, 0.5% and 0.8% respectively). *Methanoculleus marisnigri* has been reported for the ability to utilize hydrogen, formate, ethanol and a variety of secondary alcohols such as propanol and butanol, as electron donors for methanogenesis [24,25]. This metabolic feature of *Methanoculleus* makes it more tolerant to changes in the environmental conditions and therefore found in dominant in TP-EBD. The hyperthermophilic and anaerobic bacterium *Thermotoga maritima* is capable to ferment a wide variety of carbohydrates to produce acetate, $CO_2$, and $H_2$ through a classical Embden-Meyerhof pathway [26]. Similarly, *Kosmotoga* is a hydrogen producer and can grow over a wide temperature range (20–80 °C) and 20–60 g/L NaCl concentration [27,28]. The dominance of both hydrogen producer *Thermotogae* and *Kosmotaga* along with hydrogentrophic methanogen *Methanoculleus* indicated the dominance of hydrogenotrophic methanogenesis pathway at thermophilic condition. This is also supported by functional diversity analysis of these samples. As showed in the Table 3 abundance of intermediate enzyme of aceticlastic methanogenesis pathway viz. Phosphate acetyltransferase, N5-methyltetrahydromethanopterin: coenzyme M methyltransferase, Methyl coenzyme M reductase etc decreased in TP-EBD sample as compared to SM-EBD and MP-EBD while abundance of coenzyme F420-reducing hydrogenase (essential intermediate in methanogenesis from $H_2$ and $CO_2$), Acetyl-CoA acetyltransferase (involved in butyrate oxidation), Acetyl-CoA synthetase (involved in acetate formation), Formate dehydrogenase (catalyses the oxidation of formate to carbon dioxide) increased in the TP-EBD as compared to SM-EBD and MP-EBD samples. This functional diversity analysis indicated the dominance of aceticlastic methanogenesis pathway under mesophilic

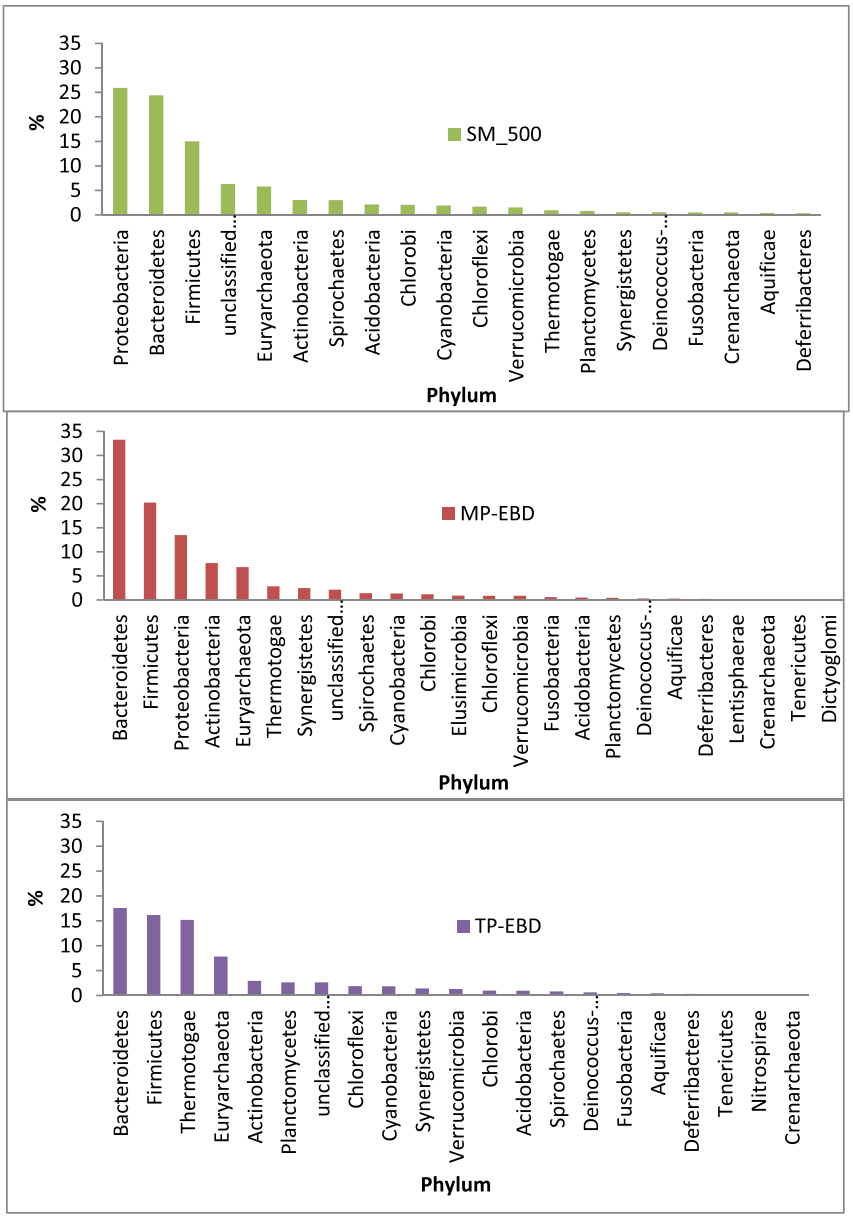



**Fig. 6.** % Normalized abundance at the Phylum level in inoculum (SM-EBD), biogas digester sample operated under mesophilic condition (MP-EBD) and under thermophilic condition TP-EBD.

condition and hydrogenotrophic methanogenesis pathway under thermophilic condition. Competition of homoacetogens and hydrogenotrophic methanogens for hydrogen at pshycrophilic condition has been reported by Kotsyurbenko et al. [29]. Homoacetogen is the dominated community at the low temperature and could not compete with hydrogenotrophic methanogens at high temperature. Members of *Treponema* (homoacetogens) has been reported to produce acetate from $H_2$ and $CO_2$ and also used as an indicator of process instability [30]. High abundance of *Treponema* sp. indicated deterioration in the digester. In this study, the % abundance of *Treponema* sp. is reduced drastically in TP-EBD (0.2%) as compared to SM-EBD (1.3%). Reduction of *Treponema* sp. abundance in TP-EBD indicated the dominance of hydrogenotrophic methanogenesis pathway and stability of digester under thermophilic condition.

In this study, the methane yield of AD under thermophilic condition is found to be higher than that of mesophilic condition, even though the free ammonia concentration is almost two times. The metagenomic analysis revealed that higher temperature not only affects the dissociation constant of ammonia, but also alters the microbial diversity of the digester, resulting in switching of methanogenesis in the digester from aceticlastic pathway to hydrogenotrophic pathway. Since, hydrogenotrophic methanogens are more resistant to high ammonia concentrations [9], therefore the digester produced more methane under thermophilic and high ammonia concentrations.

## 4. Conclusion

The results obtained in this study showed that under the optimized hydrolysis the average methane yield of thermophilic anaerobic digestion is higher, although the free ammonia level is double than that of mesophilic anaerobic digestion. In both the conditions, ammonia concentration in the slurry was found to be

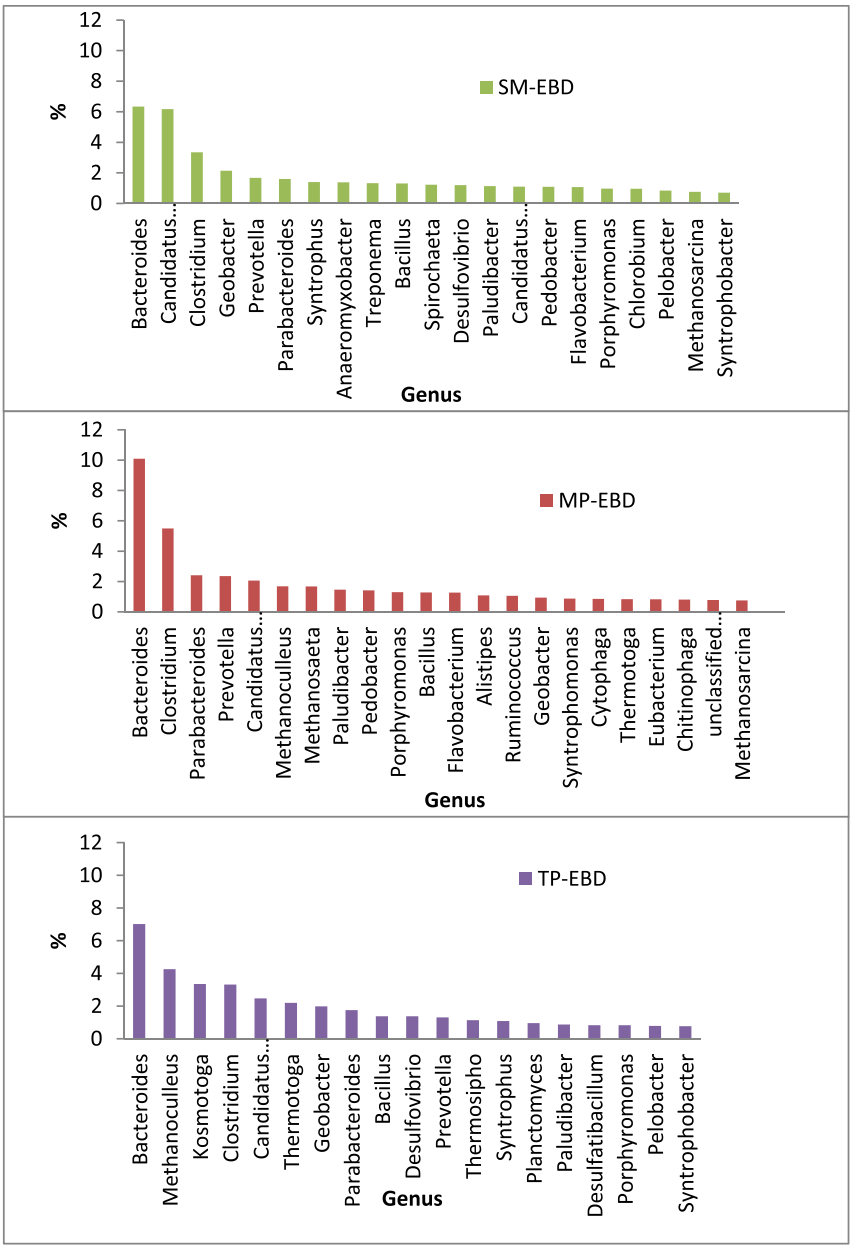



**Fig. 7.** % Normalized abundance at the Genus level in inoculum (SM-EBD), biogas digester sample operated under mesophilic condition (MP-EBD) and under thermophilic condition TP-EBD.

**Table 3**
Functional diversity in SM-EBD, MP-EBD and TP-EBD samples.

| Gene | Function | SM-EBD | MP-EBD | TP-EBD |
|---|---|---|---|---|
| Phosphate acetyltransferase (EC 2.3.1.8) | Intermediate enzyme of acetoclastic methanogenesis pathway | 0.104 | 0.088 | 0.030 |
| N5-methyltetrahydromethanopterin: coenzyme M methyltransferase | Intermediate enzyme of acetoclastic methanogenesis pathway | 0.080 | 0.056 | 0.048 |
| Methyl coenzyme M reductase | Intermediate enzyme of acetoclastic methanogenesis pathway | 0.080 | 0.088 | 0.044 |
| Coenzyme F420-reducing hydrogenase | Essential intermediate in methanogenesis from $H_2$ and $CO_2$ | 0.008 | 0.007 | 0.030 |
| Acetyl-CoA acetyltransferase (EC 2.3.1.9) | Butyrate oxidation | 0.076 | 0.088 | 0.133 |
| Acetyl-CoA synthetase | Acetate formation | 0.140 | 0.059 | 0.241 |
| Formate dehydrogenase (EC 1.2.1.2) | Catalyse the oxidation of formate to carbon dioxide | 0.220 | 0.12 | 0.204 |

below toxic levels which show that the optimized hydrolysis of KW is useful for controlling ammonia level during biomethanation. Microbial diversity analysis also showed the stability of biogas digester under thermophilic condition as *Treponema* sp. abundance was lower (0.2%) than the mesophilic condition (1.3%). The abundance of *Methanoculleus* in the thermophilic condition along with *Thermotogae* and *Kosmotoga* sp. in the TP-EBD sample revealed the shifting of aceticlastic methanogenesis pathway to

hydrogenotrophic methanogenesis pathway. Outcomes of this research could be useful for biogas production using kitchen waste where ammonia accumulation in the process is a major problem. The metagenomic analysis finding will help to develop methods for bioaugmentation of ammonia tolerant thermophilic consortia.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2019.02.023.

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